

Bounded Logical Step Sequencing

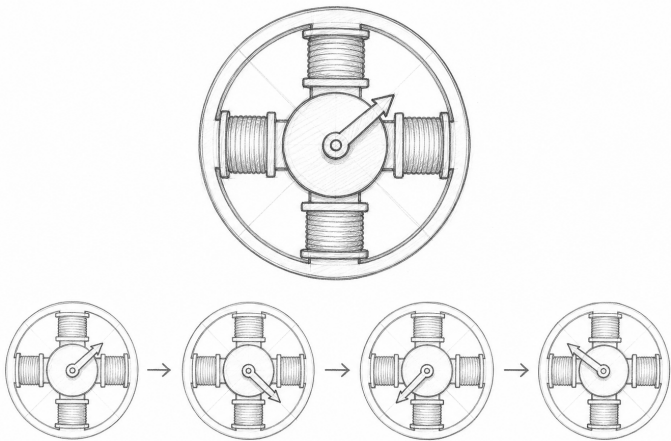
Turn copied commands into one safe coil-intent frame at a time

E0 SYNTHETIC REPLAY — NO POWERED MOTOR OR DRIVER CLAIM

Lesson	047 — component	Time	90–120 minutes
Prerequisite	status and wrap-safe time	Board	compile-only Mega replay
Evidence	named memory result cells	Boundary	no pins, timer, driver, supply, or motor

Responsibility. Convert one structurally valid logical command and caller-supplied time into a bounded four-bit coil *intent*. Advance at most one due transition per call. Expire, cancel, stop, fault, and reset to all-off intent. Logical position is not physical position, homing, travel measurement, or permission to energize coils.

1 Name the eight logical frames



Alternative text: a graphite pencil concept drawing shows a four-coil rotor and four successive pointer positions. It illustrates logical sequence order only; it is not an electrical schematic or drive contract.

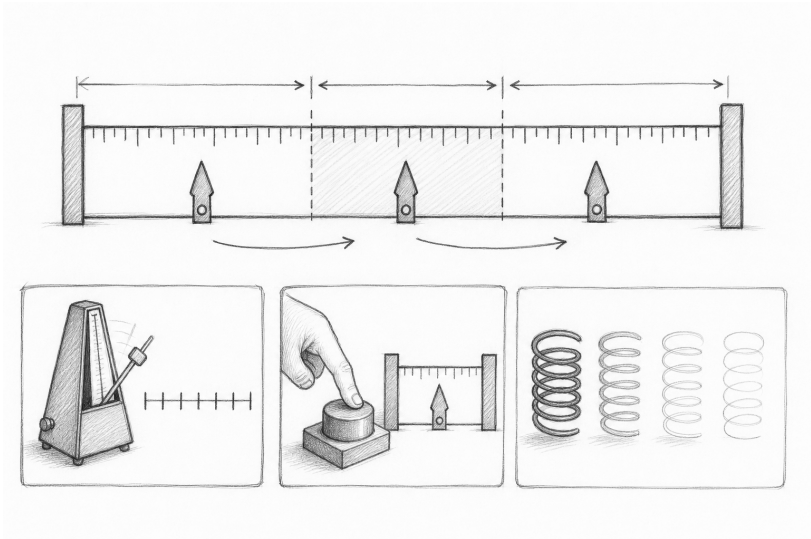
The fixed half-step sequence is

0001, 0011, 0010, 0110, 0100, 1100, 1000, 1001.

Forward walks right through this list; reverse walks left. Each bit is a logical intent only. It names neither a Mega pin nor a ULN2003 terminal.

Index	Frame	Next forward	Next reverse	Prediction note
0	0001	0011	1001	_____
2	0010	_____	_____	_____
4	0100	_____	_____	_____
6	1000	_____	_____	_____

2 Bound rate and travel before acceptance



Alternative text: a pencil storyboard shows a pointer confined between two logical limits, caller-timed ticks, cancellation by a stop gesture, and coil shapes fading to all-off intent.

Configuration fixes the minimum and maximum interval, maximum command age, minimum and maximum logical position, and whether a normally completed command retains its last frame. Reject zero or wrap-unsafe durations, reversed bounds, or a position range that widened arithmetic cannot update.

A moving command needs a nonzero ID, OK status, non-future issue time, nonzero step count, interval inside the configured range, and a complete endpoint inside the logical bounds. A stopped command requires zero steps. Structural invalidity is rejected without changing accepted state.

Start	Direction / steps	Intended endpoint	Accept / reject
0	forward / 8	_____	_____
7	forward / 2	_____	_____
-7	reverse / 1	_____	_____

One-transition timing worksheet

The first accepted moving command publishes all-off 0000. Its first deadline is *issuedAt + stepInterval*. The first logical frame appears only on a call at or after that deadline. At every due call, prepare exactly one next frame. If three deadlines elapsed, three repeated calls catch up one frame at a time; the deadline cadence is retained rather than stretched to “now.”

now (ms)	next due (ms)	before/at/after	frames advanced	next due
9	10	_____	_____	_____
10	10	_____	_____	_____
35	20	overdue	_____	_____

3 Use preview, then commit

`preview()` is side-effect free. It prepares an opaque candidate bound to one policy object and generation. `canCommit()` rejects a foreign, stale, replayed, or already-used candidate. Only `commit()` publishes the complete next snapshot. This seam lets Lesson 048 prove several candidate policies admissible before changing any of them.

1. Preview a valid command and copy the current snapshot. They must differ.
2. Commit the candidate once. Copy the new snapshot.
3. Try the same candidate again. It must be inadmissible and atomic.
4. Prepare from a second policy. The first policy must reject that foreign candidate.

Attempt	<code>canCommit</code> prediction	status	state unchanged / changed
fresh owned preview	_____	_____	_____
double commit	_____	_____	_____
foreign owner	_____	_____	_____
preview after reset	_____	_____	_____

4 Distinguish completion, cancellation, stop, and fault

Event	Published result	What may happen next
normal completion	Complete ; final frame held only when configured	a forward-ID valid command may start
copied cancel	Cancelled ; intent 0000	a forward-ID valid command may start without reset
independent <code>stop()</code>	Cancelled ; intent 0000	a forward-ID valid command may start without reset
age expiry or time fault <code>reset()</code>	Fault ; intent 0000 canonical Inactive ; intent 0000	reset() is required before new motion remains initialized; a valid explicit command may start immediately

Cancellation dominates command meaning only after structural validity is established and only for a live command. The independent stop is idempotent. Neither path proves that current stopped flowing in physical coils.

5 Replay and inspect exact result cells

Use the Lesson 047 compile-only replay. It stores copied command fields, operation status, phase, disposition, direction, logical position, completed steps, four-bit drive intent, deadline, and sequence status in named volatile memory cells. Predict first; then record exact values.

Frame	phase / disposition	position / completed	com-	intent / next due	interpretation
0 accept forward	_____	_____		_____	_____
1 due step	_____	_____		_____	_____
2 due step	_____	_____		_____	_____
3 replace reverse	_____	_____		_____	_____
4 due reverse	_____	_____		_____	_____
5 copied cancel	_____	_____		0000 /	_____
6 restart forward	_____	_____		_____	_____
7-8 finish	_____	_____		_____	_____
9 independent stop	_____	_____		0000 / none	_____

E0 observation path. Named memory result cells are the non-Serial evidence path for this pure replay. Mega compilation proves packaging only. Do not run it on a powered board: the sketch owns no endpoint, timer, interrupt, driver, supply, motor, stop switch, test point, or wiring.

6 Record measured resource evidence

Artifact	Measured	Gate	Result / interpretation
Lesson 047 Mega flash	8,068 B	16,384 B	_____
Lesson 047 static SRAM	1,053 B	2,048 B	_____
BoundedStepperSequence object	_____ B	target 128 / hard 176 B	_____
host test executable	_____ B	recorded host-size gate	_____

The first two measurements use the pinned Mega 2560 toolchain. Record the tested commit and tool versions beside any newly measured blank. Size evidence is not runtime, thermal, motion, or physical safe-state evidence.

7 Diagnose the policy boundary

Evidence	Interpretation
same ID, changed payload	rejected mutation; accepted snapshot remains intact
forward modular ID while live	admissible replacement only if its full endpoint is bounded
ID regression or half-range ambiguity	rejected without mutation
several deadlines overdue	one transition per call; repeat calls to catch up
position exactly at configured limit	admissible endpoint, never evidence of a physical stop
expired command	Fault , all-off intent, reset required
cancel or independent stop	Cancelled , all-off intent; a later forward-ID command may start
hold-at-rest completion	logical frame may remain; no E2 hold-time or temperature claim

8 Separate acquisition from safe state

Stage	Predict	Observe and interpret
Acquire	initialize valid policy configuration	status cell: _____
Act	commit one admitted logical transition	phase/position/intent: _____
Safe state	cancel, stop, expiry, fault, or reset	exact intent: 0000; status: _____

Initialization acquires no hardware resource. All-off is a software intent, not electrical high impedance or measured current removal. Resource acquisition and safe-state evidence remain separate.

9 Blank E2 promotion record

Leave every field blank at E0. Compilation, simulation, retail listings, and this workbook cannot fill a powered-motion acceptance cell.

Required E2 field	Bench acceptance record
exact motor / driver / board revisions and specimen markings	_____
authoritative formal schematic and mapped coil order	_____
separate supply rating, common reference, clamp path, measured current	_____
restraint, travel envelope, clearance, and guarded load	_____
independent stop and physical power-removal prediction / observation	_____
acquisition evidence separate from de-energized safe-state evidence	_____
maximum hold duration and winding / driver	_____
temperature evidence	_____
physical step count, direction, missed-step, and endpoint observations	_____
tool version / tested commit / reviewer / date	_____

10 Evidence links

Open the public interface, deterministic tests, compile-only replay, and design plan with open gates. The searchable HTML lesson is the hardware-independent accessible route. This untagged print edition makes no PDF/UA, powered-driver, physical-motion, thermal, restraint, or stop-circuit acceptance claim.